SEMINAR REPORT

A seminar Report on

Smart Power Management

Submitted in partial fulfillment for the award of

THIRD YEAR DIPLOMA IN

COMPUTER ENGINEERING

By

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DEPARTMENT OF COMPUTER ENGINEERING

S.S.V.P.S's Bapusaheb Shivajirao Deore Polytechnic, Dhule

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TABLE OF CONTENT

Sr. No.	Chapter	Page No.
	Abstract	i
	List of Figures	ii
	List of Tables	iii
1	Introduction	1
2	Literature Review	2
3	Information/Chapters	3
4	Advantages and Disadvantages	8
5	Conclusion	9
6	Project Initiation :	10
	6.1) Description of problem statement.	10
	6.2) Scope and objectives.	10
	6.3) Stake holder	11
	6.4) Platform/ Equipment/ Resources identification.	12
7	Seminar Presentation	
	Bibliography	
	References	

ABSTRACT

The increasing demand for electrical energy and the wastage of power in residential, commercial, and industrial environments create significant challenges in energy management. Conventional power management systems often lack real-time monitoring and intelligent control, which can lead to unnecessary energy consumption, higher electricity bills, and inefficient utilization of electrical resources.

To address these challenges, this project presents a Smart Power Management System, designed to monitor, control, and optimize electrical energy consumption efficiently. The system uses smart sensors, monitoring devices, and automated control mechanisms to collect real-time power consumption data and identify unnecessary energy usage. Based on the collected information, the system helps users manage electrical appliances, reduce power wastage, and improve energy efficiency. It can also provide consumption reports and alerts to support better decision-making. The proposed system aims to promote energy conservation, reduce electricity costs, and support the efficient utilization of available power resources. Smart Power Management offers a practical approach toward sustainable energy usage in homes, offices, and industries.

LIST OF FIGURES

Fig. No.	Title	Page No.
1.1	Working of AI Privacy Gateway	4
1.2	Architecture of AI Privacy Gateway	6

LIST OF TABLES

Table No.	Title	Page No.
2.1	Literature Review	2
6.1	Resources	12

CHAPTER 1
INTRODUCTION

Electrical energy is widely used in residential, commercial, and industrial environments to operate appliances, machines, lighting systems, and electronic devices. However, increasing electricity demand and unnecessary power consumption have created challenges in energy conservation and efficient power utilization. Traditional power management methods often depend on manual monitoring, which makes it difficult to track energy usage and identify power wastage in real time.

The Smart Power Management System is an energy management solution designed to monitor, control, and optimize electrical power consumption. It helps users understand how electricity is being utilized and supports the efficient operation of electrical appliances. By using suitable sensors, monitoring devices, and automated control mechanisms, the system can collect power consumption data and provide useful information about energy usage.

The proposed system focuses on improving energy efficiency through continuous monitoring and intelligent control. For example, in a home or office, the system can help identify appliances that consume unnecessary electricity and support switching off unused devices through suitable control mechanisms. This approach aims to reduce power wastage, improve energy utilization, and help users make informed decisions about electricity consumption.

The system can incorporate technologies such as smart energy meters, voltage and current sensors, micro-controllers, automated switching, and data monitoring software, depending on the implementation. The collected information can be used to monitor energy consumption, identify abnormal usage patterns, generate reports, and control connected appliances where supported. These features help make power management more convenient and efficient.

The main objective of Smart Power Management is to promote energy conservation, reduce unnecessary electricity consumption, and improve the utilization of available electrical resources. Its effectiveness can be evaluated through parameters such as energy savings, monitoring accuracy, response time, operating cost, and system reliability. Thus, Smart Power Management provides a practical approach to efficient energy usage in homes, offices, and industries.

CHAPTER 2
LITERATURE REVIEW

A Smart Power Management – Part 1

1. A Smart Home Energy Management System Using IoT and Big Data Analytics Approach

This paper presents a Smart Home Energy Management System that uses the Internet of Things (IoT) and Big Data analytics to monitor and control electricity consumption. Household appliances are connected through data-acquisition modules that collect energy usage information and transmit it to a central server. Users can monitor appliance usage remotely and switch devices ON or OFF. It can be applied to residential, commercial, and industrial environments.

2. Design and Implementation of Smart Home Energy Management System for Indian Residential Sector

This paper presents an IoT-based Home Energy Management System designed specifically for Indian households. It monitors electricity consumption and categorizes appliances according to their importance and flexibility. Non-essential appliances can be scheduled during periods of lower electricity demand or cheaper tariffs. The study reports potential electricity-cost savings of up to 49% under a fixed tariff and 53% under a Time-of-Day tariff. The proposed approach helps reduce peak demand, manage electricity costs, and improve energy efficiency.

3. State-of-the-Art and Real-Time Implementation of an IoT-Based Home Energy Management System for a Cluster of Dwellings

This paper presents a real-time IoT-based energy management system designed for individual homes and groups of homes. It uses smart plugs, a load scheduler, and a mobile application to monitor and control electrical appliances. Flexible appliances are scheduled during off-peak hours to reduce electricity costs and peak demand while maintaining user convenience. The study reports a 12.14% reduction in electricity cost and a 52.54% reduction in the peak-to average ratio for a single dwelling compared with unscheduled loads. It demonstrates the practical benefits of automated load scheduling and smart energy monitoring.

Smart Power Management – Part 2

1. IoT-Based Smart Energy Monitoring and Management

This research area focuses on using IoT technology to monitor electricity consumption in real time. Smart sensors and energy meters collect data from connected electrical appliances and transmit it to a monitoring system. Users can observe energy usage, identify appliances consuming excessive electricity, and take corrective actions. This approach supports efficient energy utilization and helps reduce unnecessary power wastage.

2. Intelligent Load Scheduling for Energy Efficiency

Intelligent load scheduling focuses on operating electrical appliances at suitable times according to energy demand, electricity tariffs, and user requirements. Essential appliances can continue operating when needed, while flexible loads can be shifted to off-peak periods. This approach can reduce peak electricity demand, improve load distribution, and help lower electricity costs.

3. Real-Time Monitoring and Automated Appliance Control

Real-time monitoring and automated appliance control use sensors, connected devices, and controllers to measure and manage electricity consumption. Users can monitor appliance status and control connected equipment remotely. Automated switching can help prevent unnecessary operation of appliances and improve convenience. These functions are important for developing efficient Smart Power Management systems.

Smart Power Management

1. Energy Conservation Through Real-Time Power Monitoring

Real-time power monitoring helps users understand how electricity is consumed by different appliances. Smart meters and sensors collect consumption data, which can be analyzed to identify high-energy-consuming devices. Users can then modify their usage patterns and reduce unnecessary electricity consumption. This method supports energy conservation and better power management.

2. Smart Energy Management Through Load Categorization

Load categorization classifies electrical appliances according to their importance and flexibility. Essential loads operate according to user requirements, whereas flexible loads can be scheduled for suitable periods. This technique helps manage peak demand and improve energy efficiency. It is particularly useful for residential energy management under different electricity tariff structures.

3. Peak Load Reduction Through Automated Scheduling

Automated scheduling manages the operating time of selected electrical appliances to reduce peak electricity demand. A scheduling system can shift flexible loads to off-peak hours while considering user convenience and appliance requirements. This approach can reduce electricity costs, improve load balancing, and support more reliable power utilization.

Smart Power Management and Energy Optimization – Summary

1. Smart Home Energy Management Using IoT

IoT-based Smart Home Energy Management connects electrical appliances to a monitoring and control system. Sensors collect energy consumption information and send it to a central server or compatible platform. Users can monitor appliances remotely and control supported devices. This approach provides better visibility of electricity usage and helps minimize unnecessary power consumption.

2. Electricity Cost Optimization Using Load Scheduling

Electricity cost optimization uses appliance scheduling and consumption analysis to manage power usage efficiently. Flexible appliances can be operated during periods when electricity demand or applicable tariffs are lower. The technique can help reduce electricity bills and peak loads without interrupting essential activities.

3. Smart Power Management for Multiple Homes

Smart Power Management can be extended from a single household to a group of homes. Connected monitoring devices and scheduling systems coordinate electricity consumption across multiple dwellings. This can help reduce combined peak demand and improve the management of electrical loads. Such systems demonstrate the potential of IoT-based energy management for larger residential environments.

Table No 2.1: Literature review

Sr.no	Approach Type	Technology Used	Dataset	Key Objective	Limitations
1	IoT-Based Smart Home Energy Management	IoT, Big Data Analytics, Wireless Communication, Data-Acquisition Modules	Residential HVAC equipment used as a laboratory case study	To monitor household energy consumption, enable remote appliance control.	Requires reliable network connectivity, additional hardware, data storage, and suitable security measures.
2	Smart Home Energy Management for Indian Residential Sector	IoT, Programmable Load Categorization, Load Scheduling, Time-of-Day Tariffs	Indian residential energy consumption and load-management analysis	To reduce electricity costs and peak demand by categorizing appliances and scheduling flexible loads.	Results depend on appliance usage patterns, tariff structure, accurate load information, and user acceptance.
3	Real-Time IoT-Based Home Energy Management for Multiple Dwellings	IoT, Smart Plugs, Load Scheduler, Mobile Application, Multi-Objective.	Hardware-based smart plug prototype and a cluster of 50 homes	To reduce electricity costs and peak-to-average ratio while maintaining user convenience.	Requires connected hardware and accurate consumption data; optimization and scheduling can be complex.
4	Smart Building Energy Optimization	IoT, AI, Smart Meters, Solar Energy, ThingWorx	Building energy consumption data	To monitor and optimize electricity usage in smart buildings.	Requires reliable sensors, connectivity, and accurate forecasting.
5	Fuzzy Logic-Based Power Management	Fuzzy Logic Controller, Arduino Mega 2560, Proteus, Batteries	Power availability and energy-source conditions	To select suitable energy sources and improve battery life.	Depends on available energy sources and proper controller design.
6	AI-Based Home Energy Management	AI, IoT, Solar PV, Energy Storage, Forecasting	Home energy consumption and demand patterns	To optimize energy usage through prediction and appliance scheduling.	Computational complexity, cybersecurity risks, and limited real-world testing.

7	ML-Based Smart Home Energy Management	LSTM, Machine Learning, Smart Meters, Cloud Computing	Household appliance energy data recorded at 1-minute intervals for 350 days.	To predict household energy consumption and improve energy efficiency.	Prediction errors and dependence on data quality.
8	IoT and Big Data-Based Energy Management	IoT, J48 Algorithm, Weka, Apache Mahout, RuleML	Smart home energy consumption and user behavior data.	To classify energy usage and recommend energy-saving actions while maintaining safety.	Security concerns and challenges in managing large amounts of data.
9	Reinforcement Learning-Based Energy Management	Q-Learning, Artificial Neural Network (ANN), Solar PV, Energy Storage System	Simulated data for a home with solar PV, an air conditioner, a washing machine.	To schedule appliances and battery operation to reduce electricity bills while maintaining comfort.	Tested through simulation for a single home; performance depends on comfort settings
10	Occupancy-Based Power Management	Raspberry Pi, Python, CNN, Sensors, Relays	Room occupancy and sensor readings	To automatically control appliances and reduce electricity wastage.	Sensor errors and privacy concerns.
11	IoT-Based Solar Energy Management	IoT, Solar PV, Sensors, Cloud Computing	Solar voltage, current, and energy-output data	To monitor solar power and improve energy management.	High initial cost, connectivity issues, and security risks.
12	IoT-Based Energy Monitoring	Smart Meters, NodeMCU, Wi-Fi, Cloud Software	Real-time and historical electricity consumption data	To monitor electricity usage and identify energy wastage.	Depends on connectivity and suitable hardware.

CHAPTER 3
INFORMATION/CHAPTERS

1. Overview of Smart Power Management:

Smart Power Management is a system designed to monitor, control, and optimize electrical energy consumption. It uses technologies such as smart sensors, energy meters, and automated control systems to track power usage and reduce unnecessary electricity consumption. Its main goal is to improve energy efficiency, reduce electricity costs, and ensure better utilization of electrical resources.

2. Electrical Energy Monitoring:

Electrical energy monitoring involves measuring and analyzing the electricity consumed by different appliances and equipment. Smart meters and sensors collect power consumption data, helping users identify high-energy-consuming devices and understand their usage patterns. This information supports better energy management and helps minimize power wastage.

3. Need for Intelligent Power Management:

Traditional power management methods often depend on manual monitoring and control, which can lead to unnecessary electricity consumption. Smart Power Management uses real-time monitoring and automated control to manage electrical loads efficiently. It helps reduce peak demand, optimize appliance operating schedules, lower electricity bills, and promote energy conservation.

1. Working Principle:

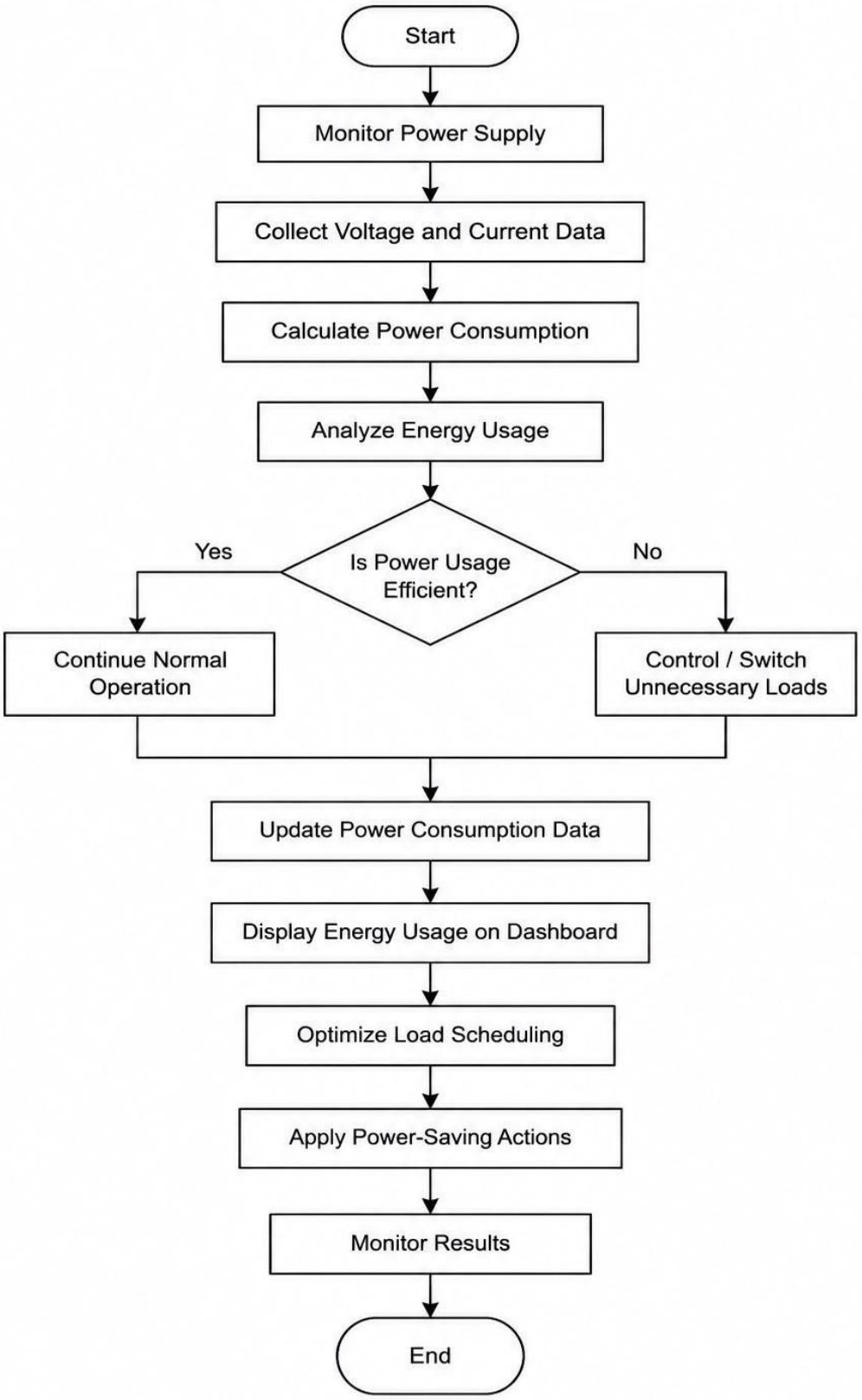


Fig 3.1:Working of Smart Power Management

The working of the Smart Power Management System consists of the following steps:

1. Start: The process begins when the Smart Power Management System is switched on and starts monitoring the electrical power supply.

2. Monitor Power Supply: The system monitors the electrical supply to track the operation of connected appliances and electrical equipment.

3. Collect Voltage and Current Data: Sensors or smart meters collect voltage and current readings from the electrical system.

4. Calculate Power Consumption: The collected readings are used to calculate power consumption and understand the amount of electricity being used.

5. Analyze Energy Usage: The system analyzes the collected data to identify unnecessary electricity consumption and high-energy-consuming appliances.

6. Decide Whether Power Usage Is Efficient: The system checks whether power consumption is within the defined limit. If usage is efficient, normal operation continues. If consumption exceeds the limit, corrective action may be required.

7. Control Electrical Loads: Based on the analysis, the system can switch off unnecessary appliances or adjust selected electrical loads through suitable control mechanisms.

8. Update Power Consumption Data: The latest energy consumption information is recorded in the system database for monitoring and future analysis.

9. Display Energy Usage: The collected information is displayed on a dashboard or mobile application, allowing users to monitor electricity consumption.

10. Generate Reports and Alerts: The system can generate energy consumption reports and alerts when excessive power usage is detected, if these features are implemented.

11. Apply Power-Saving Actions: Users or the automated control system take suitable actions to reduce unnecessary electricity consumption and improve energy efficiency.

12. End: The process completes after the power consumption data is updated and the monitoring results are displayed.

2. System Architecture and Main Components:

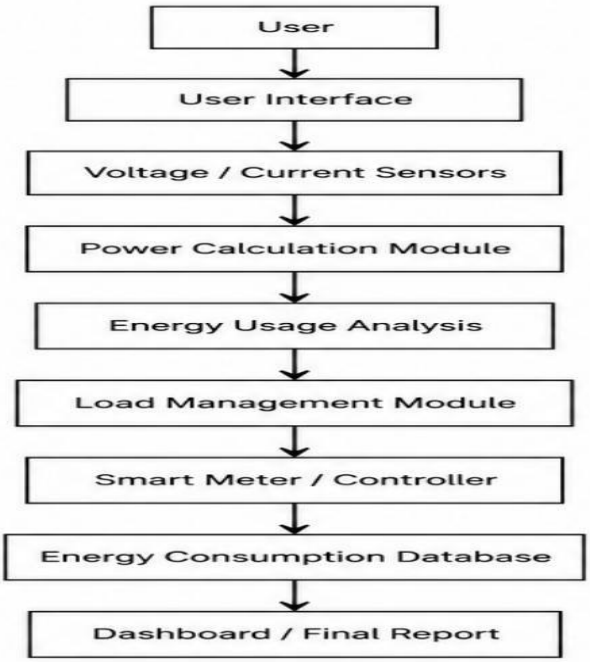


Fig 3.2: Architecture of Smart Power Management

The main components of the Smart Power Management System include the User Interface, Voltage and Current Sensors, Power Calculation Module, Energy Usage Analysis Module, Load Management Module, Smart Meter or Controller, and Energy Consumption Database. These components work together to monitor electricity consumption, analyze power usage, control electrical loads, and display energy consumption reports.

1. Technologies and Techniques Used:

Smart Power Management can be developed using technologies such as the Internet of Things (IoT), smart energy meters, voltage and current sensors, microcontrollers, and monitoring software. Sensors measure electrical parameters, while the power calculation module estimates energy consumption. Automated control mechanisms help manage connected appliances, and database systems store consumption records for analysis. The final technology selection depends on the project implementation and available resources.

2. Example of Power Consumption Monitoring:

Consider a home where a fan, television, air conditioner, and lighting system consume electricity. The Smart Power Management System monitors their power consumption and identifies appliances that use excessive electricity. If an appliance remains switched on unnecessarily, the system can notify the user or switch it off through an automated controller, if supported. This helps reduce power wastage and encourages efficient electricity usage.

3. Energy Monitoring and Power Control:

The system collects electricity consumption data through sensors or a smart meter and analyzes the readings to identify inefficient usage. The results are displayed on a dashboard or mobile application, allowing users to monitor their energy consumption. If excessive power usage is detected, the system can generate alerts or control selected appliances through suitable control mechanisms. This process helps improve energy efficiency, reduce electricity costs, and support better power management.

CHAPTER 4
ADVANTAGES AND DISADVANTAGES

4.1 Advantages:

1. Reduced Power Consumption: Helps reduce unnecessary electricity usage through continuous monitoring and efficient control of electrical appliances.
2. Improved Energy Efficiency: Analyzes power consumption and supports better utilization of electrical resources.
3. Lower Electricity Bills: Helps users identify high-energy-consuming appliances and reduce electricity costs.
4. Real-Time Monitoring: Allows users to monitor electricity consumption using smart meters, sensors, or monitoring applications.
5. Automated Load Management: Supports automatic control and scheduling of electrical appliances to reduce power wastage and peak demand.

4.2 Disadvantages:

1. High Initial Cost: Installing smart meters, sensors, controllers, and other required equipment may increase the initial cost.
2. Technical Complexity: Designing and configuring the system requires appropriate technical knowledge and skills.
3. Network Dependency: IoT-based systems may require a stable internet or communication connection for remote monitoring.
4. Maintenance Requirements: Sensors, meters, and control devices require regular maintenance to ensure reliable operation.
5. Security Risks: Connected devices and energy monitoring systems may face cybersecurity risks if proper security measures are not implemented.

CHAPTER 5
CONCLUSION

The Smart Power Management System provides an efficient approach to monitoring, controlling, and optimizing electrical energy consumption. It collects power usage data, analyzes energy consumption patterns, and helps identify unnecessary electricity usage.

By combining smart sensors, energy monitoring, load management, and automated control, the system aims to improve energy efficiency and reduce electricity costs. Although challenges such as initial installation costs, maintenance requirements, network dependency, and technical complexity remain, proper design and implementation can improve its effectiveness.

In conclusion, Smart Power Management can serve as a useful solution for conserving electrical energy, reducing power wastage, and improving the utilization of available resources in residential, commercial, and industrial environments.

CHAPTER 6
PROJECT INITIATION

a) Description of Problem Statement:

The increasing demand for electrical energy and the wastage of power in residential, commercial, and industrial environments create significant challenges in energy management. Conventional power management systems may lack real-time monitoring and intelligent control, which can lead to unnecessary electricity consumption, higher electricity bills, and inefficient utilization of electrical resources.

Traditional methods of power management often depend on manual monitoring and control of electrical appliances. This makes it difficult to identify unnecessary power consumption and manage electrical loads efficiently. Therefore, there is a need for a system that can monitor electricity usage, analyse consumption patterns, and control electrical loads to reduce power wastage.

The proposed Smart Power Management System aims to address this problem by using smart sensors, energy monitoring devices, and automated control mechanisms. It collects power consumption data, analyses energy usage, and helps users identify unnecessary electricity consumption. The objective is to improve energy efficiency, reduce electricity costs, and promote the effective utilization of available electrical resources.

b) Scope of the Project:

The scope of the Smart Power Management System is to develop a system that monitors, analyses, and helps control electrical energy consumption. The system can use smart sensors and monitoring devices to collect real-time power consumption data and identify high-energy-consuming appliances. It can support load management, automated control, and energy consumption reporting to help users make better decisions about electricity usage.

The project focuses on reducing unnecessary power consumption, improving energy efficiency, and supporting effective power management in residential, commercial, and industrial environments.

Objectives of the Project:

1. To monitor electrical energy consumption using smart sensors or monitoring devices.

2. To collect and analyse power consumption data.

3. To identify unnecessary electricity usage and high-energy-consuming appliances.

4. To improve energy efficiency through effective power management.

5. To support load scheduling and automated control of electrical appliances where applicable.

6. To reduce electricity costs and promote energy conservation.

c) Stakeholders:

The stakeholders of the Smart Power Management System are the individuals and groups who use, develop, maintain, or benefit from the system.

1. **End Users:** Individuals who use the system to monitor electricity consumption and reduce power wastage in homes, offices, or other buildings.

2. **Developers:** The team responsible for designing, developing, testing, and maintaining the Smart Power Management System.

3. **Electricity Service Providers:** Organizations that supply electrical energy and may benefit from improved energy utilization and reduced peak demand.

4. **System Administrators:** Individuals responsible for managing the system, monitoring its performance, and maintaining its proper operation.

5. **Organizations:** Residential societies, educational institutions, businesses, and industries that want to improve energy efficiency and control electricity costs.

6. **Project Guide and Evaluators:** Faculty members who guide the development process and evaluate the project's design, implementation, and results.

d) Platform / Equipment / Resources:

1. Platform:

The Smart Power Management System can be developed on the Windows or Linux operating system using suitable programming languages and tools. A code editor such as Visual Studio Code can be used for development and testing. The system may use smart sensors, energy meters, and a microcontroller to monitor power consumption and control electrical appliances. A database can be used to store energy consumption data and display usage reports.

2. Equipment:

- **Computer or Laptop:** For developing and testing the application.
- **Processor:** A standard modern processor suitable for Python development.
- **RAM:** Minimum 8 GB recommended; additional memory may be required for running local language models.
- **Internet Connection:** Required for accessing external AI services and downloading libraries.
- **Storage:** Sufficient space for source code, required libraries, and model files, if used.

3. Resources:

Resources	Name	Purpose
Operating System	Windows 11	Host and run development tools, editors, and local testing environment.
Programming Language	Python 3.11	Primary language used to implement application code, scripts, and tests.
Code Editor	Visual Studio Code	Code editing, debugging, and extension support for development.
Iot Development Tool	Arduino IDE	Used to write, upload, and test programs on compatible microcontrollers.
Sensor Data Processing	Python Libraries	Used to process voltage, current, and power consumption data collected from sensors or monitoring devices.
Database	MySql	Used to store, organize, and retrieve electricity consumption records and related system data.
Data Visualization	Matplotlib	Used to display energy consumption patterns and generate graphical reports.

CHAPTER 7
SEMINAR PRESENTATION

S.S.V.P.S's B. S. Deore Polytechnic, Dhule
Department of Computer Engineering
Academic Year : 2026-2027

SEMINAR AND PROJECT INITIATION COURSE

Sem: CO 5 K	SPI – 315003	Date:

Topic: **Smart Power Management**

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Contents

- Abstract
1. Introduction
2. Literature Review
3. Information to Seminar Topic
4. Advantages and Disadvantages
5. Conclusion
6. Project Initiation:
   - 6. 1 Description of Problem Statement
   - 6. 2 Scope and Objectives
   - 6. 3 Stakeholder
   - 6. 4 Seminar Discussion Plan
   - 6. 5 Platform/ Equipment/ Resources Identification.
1. Bibliography
2. References

Abstract

Smart Power Management is an efficient approach to monitoring and controlling electricity consumption. Traditional power management systems may lead to unnecessary energy wastage due to a lack of real-time monitoring and automated control. The proposed system uses sensors, microcontrollers, smart meters, and IoT technology to monitor power consumption and optimize electricity usage.

It aims to reduce energy wastage, lower electricity costs, and improve energy efficiency, promoting sustainable power utilization.

3

1. Introduction

- Smart Power Management is a technology-driven approach to monitor and control electricity consumption efficiently.
- It uses sensors, microcontrollers, smart meters, and IoT technology to track power usage.
- It helps manage electrical appliances and reduce unnecessary energy wastage.
- The system aims to lower electricity costs and improve energy efficiency.
- It promotes sustainable and responsible utilization of electrical power.



Smart Power Management Cycle

4

2. Literature Survey

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
1	IoT and Big Data Analytics	IoT, wireless communication, data acquisition modules, centralized server, Big Data analytics	Residential HVAC energy consumption data	To monitor electricity consumption and control appliances remotely.	Requires reliable internet, additional hardware, data storage and security.
2	IoT-Based Load Management	IoT, programmable load categorization, appliance scheduling, Time-of-Day tariffs	Indian residential electricity consumption and usage patterns	To reduce peak electricity demand and electricity bills through smart load scheduling.	Savings depend on appliance usage, electricity tariffs and user acceptance.
3	Real-Time IoT-Based Energy Management	IoT, smart plugs, load scheduling, mobile application, MMGO optimization algorithm	Energy consumption data from a single dwelling and a cluster of 50 homes	To reduce electricity costs and peak demand while maintaining user convenience.	Requires connected hardware, accurate energy data and complex scheduling algorithms.

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
4	ML-Based Smart Home Energy Management	LSTM, Machine Learning, Smart Meters, Cloud Computing	Household appliance energy data recorded at 1-minute intervals for 350 days.	To predict household energy consumption and improve energy efficiency.	Prediction errors and dependence on data quality.
5	IoT and Big Data-Based Energy Management	IoT, J48 Algorithm, Weka, Apache Mahout, RuleML.	Smart home energy consumption and user behavior data.	To classify energy usage and recommend energy-saving actions while maintaining comfort and safety.	Security concerns and challenges in managing large amounts of data.
6	Reinforcement Learning-Based Energy Management	Q-Learning, Artificial Neural Network (ANN), Solar PV, Energy Storage System	Simulated data for a home with solar PV, an air conditioner, a washing machine, and energy storage.	To schedule appliances and battery operation to reduce electricity bills while maintaining comfort.	Tested through simulation for a single home; performance depends on comfort settings and operating conditions.

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
7	AI-Based Sustainable Energy Management	Machine Learning, Deep Learning, Energy Forecasting	Data from over 200 reviewed research studies	To predict electricity demand and improve energy efficiency.	Data quality issues and limited real-world implementation.
8	AI and IoT Home Energy Management	AI, IoT, Solar PV, Energy Storage	Household energy consumption data	To optimize appliance scheduling and reduce electricity costs.	Cybersecurity risks and device compatibility issues.
9	Machine Learning-Based Energy Management	IoE, SVM, Cloud Storage, Data Preprocessing	757 energy-management records	To predict energy-management requirements and reduce energy wastage.	Requires computing resources and further real-world testing.

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
10	Occupancy-Based Power Management	Raspberry Pi, Python, CNN, Sensors, Relays	Room occupancy and sensor readings	To automatically control appliances and reduce electricity wastage.	Sensor errors and privacy concerns.
11	IoT-Based Solar Energy Management	IoT, Solar PV, Sensors, Cloud Computing	Solar voltage, current, and energy-output data	To monitor solar power and improve energy management.	High initial cost, connectivity issues, and security risks.
12	IoT-Based Energy Monitoring	Smart Meters, NodeMCU, Wi-Fi, Cloud Software	Real-time and historical electricity consumption data	To monitor electricity usage and identify energy wastage.	Depends on connectivity and suitable hardware.

3. Information to Seminar Topic

- **Technology:** Internet of Things (IoT), Sensors, Microcontrollers and Smart Energy Monitoring.
- **Main purpose:** To monitor and manage electricity consumption efficiently and reduce unnecessary energy wastage.
- **Key concept:** Real-time monitoring of power usage and efficient control of electrical appliances.
- **Approach:** Uses sensors, smart meters and microcontrollers to monitor electricity consumption and support better energy management.
- **Main benefit:** Reduces energy wastage, lowers electricity costs and improves energy efficiency.
- **Applications:** Smart homes, offices, colleges, industries, hospitals and commercial buildings.

9

4. Advantages / Disadvantages

Advantages:

- Reduces unnecessary electricity consumption.
- Helps lower electricity bills.
- Monitors power usage in real time.
- Supports automatic control of electrical appliances.
- Promotes energy conservation and sustainability.

Disadvantages:

- Requires high initial installation costs.
- Depends on internet connectivity for IoT features.
- Requires regular maintenance.
- May face cybersecurity and data privacy risks.
- Requires technical knowledge for installation and troubleshooting.

10

5. Conclusion

Smart Power Management is an effective approach to monitoring and managing electricity consumption. It uses smart technologies to identify energy usage, reduce unnecessary wastage, and improve the efficiency of electrical appliances. It also helps lower electricity bills and supports better power utilization. Overall, Smart Power Management promotes energy conservation, reduces environmental impact, and contributes to a smarter and more sustainable future.



11

6. Project Initiation

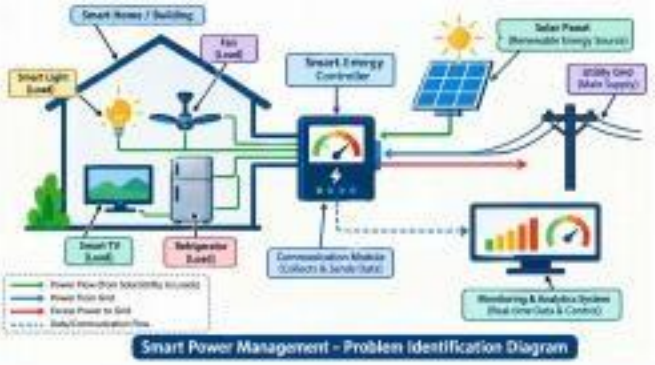
The Smart Power Management project was developed to address the problem of unnecessary electricity consumption and energy wastage in homes, offices, and other buildings. Traditional power management methods may not provide efficient real-time monitoring and control of electricity usage.

The project aims to monitor power consumption using smart meters, sensors, microcontrollers, and IoT technology. It helps identify unnecessary energy usage and supports efficient management of electrical appliances. As a result, the system aims to reduce electricity bills, improve energy efficiency, and promote energy conservation for a smarter and more sustainable future.

12

6. 1 Description of Problem Statement. Identification.

Smart Power Management aims to address electricity wastage caused by the lack of real-time monitoring and efficient control of power consumption. Traditional methods make it difficult to identify unnecessary energy usage and manage appliances effectively. The proposed system focuses on monitoring electricity consumption, identifying wastage, and improving energy efficiency.



13

6. 2 Scope and Objectives

Objectives:

- Monitors electricity consumption in real time.
- Identifies unnecessary power usage and reduces energy wastage.
- Improves the efficiency of electrical appliances.
- Reduces electricity bills through effective power management.
- Promotes energy conservation and sustainable electricity usage.

Scope:

- Applicable in homes, offices, colleges, and industries.
- Real-time monitoring of power consumption using smart meters and sensors.
- Integration of microcontrollers and IoT technology for efficient power management.
- Remote monitoring and automatic appliance control, where implemented.
- Future integration with renewable energy sources and smart grid technologies.

14

6. 3 Stakeholder

- **Homeowners:**
  Monitor electricity usage and reduce electricity bills.
- **Building Managers:**
  Manage power consumption in offices, colleges, and commercial buildings.
- **Electricity Providers:**
  Analyze energy consumption and support efficient power distribution.
- **Industries:**
  Reduce energy wastage and improve operational efficiency.
- **System Administrators:**
  Maintain the system, monitor performance, and manage connected devices.

15

6. 4 Seminar Discussion Plan

1. Current Scenario: Increasing electricity consumption creates a need for smart power management systems.

2. Energy Challenges: Uncontrolled power usage causes energy wastage and higher electricity bills.

3. Existing Approaches: Smart meters and IoT-based systems help monitor and control electricity consumption.

4. Research Gap: Efficiently identifying and reducing unnecessary power consumption remains a challenge.

5. Proposed Solution: The proposed system monitors electricity usage and helps minimize power wastage.

6. Expected Impact: Better power management can reduce costs and improve energy efficiency.

7. Future Work: AI prediction and renewable energy integration can make the system more efficient.

16

6.5 Platform/ Equipment/ Resources

1. Development Platform

- Windows-based computer for system development.
- Programming environment for processing and analyzing energy data.
- IoT platform for monitoring electricity consumption.

2. System Equipment

- Sensors and smart meters to collect power usage data.
- Arduino or ESP32 microcontroller to process sensor readings.
- Wi-Fi module or connectivity for transferring data.

3. Supporting Resources

- Electrical appliances for system testing.
- Sample energy consumption data for analysis.
- Testing tools to evaluate monitoring accuracy and system performance.

17

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19

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20

CHAPTER 8
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SEMINAR and PROJECT INITIATION Assessment / Evaluation

Sr. No	Enrollment Number	Seat Number	Name of student
1	24610960193	388511	Nagarale Dakshyani Kiran
2	24610960201	388514	Harwani Aryan Shankar
3	24610960202	388515	Jadhav Chaitanya Suresh
4	24610960206	388518	Kamble Meghana Manoj

Formative Assessment

Sr. No.	Criteria	Weightage	Exam Seat Number			
			388511	388514	388515	388518
1	Selection Topic / Theme of seminar	5				
2	Literature review and data presentation	5				
3	Quality of Preparation and innovativeness	5				
4	Q-A handling	5				
5	Time Management	5				
6	Seminar Presentation report	10				
7	Selection of Theme of Problem Statement and its innovativeness	5				
8	Stages of development of Action plan	5				
9	Prototyping	5				
	Total	**50**				
	Scaled to 25					

Sign :

Name : ----------

(Course Expert/s)